

Leases

Many companies, especially retailers and airlines, lease their assets from other companies rather than purchasing the assets outright. They do this for many reasons, including greater flexibility and to lower taxes.

In the past, clever use of accounting rules allowed companies to keep assets and debts off balance sheets. These included leased assets and their corresponding debts, securitized assets like receivables, and unfunded retirement obligations. In some cases, this helped companies manage cash flow or take advantage of alternative routes to raise funds. In other instances, off-balance-sheet items were used to artificially boost results such as earnings per share or return on assets.

In response, the International Accounting Standards Board (IASB) and the Financial Accounting Standards Board (FASB) made significant changes to their guidelines. As of 2019, companies are required to capitalize nearly all asset leases, including operating leases, on their balance sheet.¹ This stands in stark contrast to past guidelines, where a company could rent an asset, even for long periods, and recognize only the periodic rental expense.

The new accounting guidelines bring the treatment of operating leases closer to the underlying principles of this book. Implementation of the new guidelines, however, differs across accounting bodies, so incorporating operating leases into your valuation still requires special care.

This chapter begins with a review of the new accounting rules, how they differ across accounting bodies, and how they are presented on the financial statements. We then outline how to incorporate operating leases into an enterprise valuation. Since operating leases affect each part of the valuation, this chapter provides a review of the valuation principles outlined in Part Two. As companies will not revise their historical financial statements, we discuss how

¹ The International Accounting Standards Board (IASB) published IFRS 16, “Leases,” in January 2016, and the Financial Accounting Standards Board (FASB) issued Accounting Standards Update (ASU) 2016-02, “Leases (Topic 842)” in February 2016.

to adjust past financial statements to assure consistent benchmarking over time. The chapter concludes with a discussion of an alternative method for lease valuation, which can be helpful when benchmarking across companies.

ACCOUNTING FOR OPERATING LEASES

Although both IASB and FASB now require capitalization of operating leases, there are differences in implementing the new standards. For companies that use International Financial Reporting Standards (IFRS), nearly all leases greater than one year are treated as “finance” leases, meaning that leased assets and their corresponding liabilities are capitalized on the balance sheet, and lease expense is appropriately split between depreciation and interest expense. The enterprise valuation methodology outlined in Part Two of this book will correctly incorporate leases under IFRS without further adjustment.

Capitalizing leases under U.S. Generally Accepted Accounting Principles (GAAP) is more complicated. Companies classify asset leases into either “finance” leases, similar to IFRS, or “operating” leases. A lease is classified as a finance lease if cumulative payments to the lessor exceed certain thresholds.² Finance leases are not typically visible on the financial statements, as each element is embedded within another financial statement account. The leased asset is included with property, plant, and equipment. The leased liability is included with short-term and long-term debt.

To better understand the impact of the new accounting guidelines, we analyze and value FlightCo, a hypothetical airline that uses operating leases. To avoid the unnecessary complexities of continuing value, we assume the company leases only one aircraft and plans to liquidate at the end of the third year. At that point, parts inventory is sold, general obligation debt is retired, and a liquidating dividend is paid. The lease is classified as an operating lease because the contract length is significantly shorter than the life of the underlying asset.

Exhibit 22.1 presents the valuation of FlightCo’s operating lease using discounted cash flow. The lease has payments of \$9 million, \$9 million, and \$12 million in years 1 through 3, respectively. Using discounted cash flow at a cost of debt of 5 percent, the lease has a present value of \$27.1 million. Because the contract life is only three years, the present value of lease payments is significantly lower than the asset’s actual value. Later in this chapter, under “An Alternative Method for Valuing Operating Leases,” we present a valuation method for estimating the full value of leased assets using data in the annual report. This is helpful when benchmarking two companies that have different financing policies.

² A lease is classified as a finance if the lease term is greater than 75 percent of the useful life, if the present value of lease payments is greater than 90 percent of the original cost, if the asset is specialized and has no value to the lessor once returned, or if ownership of the asset is transferred to the lessee at the end of the lease.

EXHIBIT 22.1 **FlightCo: Valuation of Operating Lease**

\$ million

	Lease payment	Discount factor	Discounted cash flow
Year 1	9.0	0.952	8.6
Year 2	9.0	0.907	8.2
Year 3	12.0	0.864	10.4
Present value of operating lease			27.1

Exhibit 22.2 presents the financial statement accounts related to leases for FlightCo and shows how the accounts evolve over time. Although the evolution is neither required for valuation nor disclosed in practice, it will be helpful for explaining the accounting behind leases.

On the income statement, U.S. GAAP requires the total lease payments of \$30 million to be spread evenly over the life of the contract, even though the cash payments change over time.³ The annual lease expense is recorded on the income statement at \$10 million per year. The lease payment covers the depreciation of the asset, as well as financial compensation for the lessor.

On the balance sheet, the present value of lease payments is recorded as a “right-of-use” asset, and the corresponding liability is recorded as an “operating lease.” Both accounts start at \$27.1 million—the present value of the lease. While the two accounts will match when initially recorded, they will

EXHIBIT 22.2 **FlightCo: Financial Statement Accounts Related to Leases**

\$ million

	Year 1	Year 2	Year 3
Income statement			
Lease expense	10.0	10.0	10.0
Assets: Right-of-use asset			
Right-of-use asset, start	27.1	18.5	9.4
Lease expense	(10.0)	(10.0)	(10.0)
Embedded interest ¹	1.4	1.0	0.6
Right-of-use asset, end	18.5	9.4	—
Liabilities: Operating lease			
Lease principal, start	27.1	19.5	11.4
Interest at 5%	1.4	1.0	0.6
Lease payment	(9.0)	(9.0)	(12.0)
Lease principal, end	19.5	11.4	—

¹ Under U.S. GAAP, the interest on the operating lease liability is netted against the lease expense to determine the annual reduction in the right-of-use asset.

³ Some might think the expense accounting for operating leases mirrors that of finance leases; it does not. In a finance lease, the *present value* is straight-line amortized over the life of the lease. If FlightCo's lease were a finance lease, amortization expense would equal \$9.03 million per year for three years. Lease amortization is included in the depreciation and amortization. Lease interest expense is calculated on the liability and included in interest expense.

not match over time if cash payments vary year to year. That is the case in our FlightCo example. The right-of-use asset will decline \$8.6 million in the first year (from \$27.1 to \$18.5 million), equal to the lease expense of \$10.0 million less interest of \$1.4 million. (As if this were not confusing enough, interest is calculated on the operating lease *liability*, not the asset.) In the same year, the operating lease liability declines by \$7.6 million (from \$27.1 to \$19.5 million), equal to the cash payment of \$9.0 million less the interest of \$1.4 million.

Because of the mismatch described in the previous paragraph, most companies report operating lease liabilities that differ from their corresponding right-of-use assets. Delta Airlines, for example, reported \$6.0 billion in right-of-use assets in its 10-Q for the first quarter of 2019. In contrast, current maturities of operating leases equal \$941 million, and noncurrent operating leases equal \$5.8 billion, totaling \$6.7 billion. In this reporting period, the corresponding values differ by more than 10 percent.

VALUING A COMPANY WITH OPERATING LEASES

Incorporating operating leases into an enterprise valuation follows the same process outlined in Part Two of this book. We use four steps to value FlightCo:

1. Reorganize the financial statements. During the reorganization, adjust earnings before interest, taxes, and amortization (EBITA) upward by removing the implicit interest in operating lease expense. Adjust operating taxes to determine adjusted net operating profit after tax (NOPAT).
2. Estimate free cash flow (FCF), using adjusted NOPAT and changes in the right-of-use asset. Liabilities classified as operating leases should be treated as debt and incorporated into the reconciliation of free cash flow.
3. Estimate a weighted average cost of capital (WACC) that includes the value of the operating lease liability as debt.
4. Value the enterprise by discounting free cash flow (based on the adjusted NOPAT) at the WACC, including operating leases. Subtract traditional debt and the value of operating lease liability from enterprise value to determine equity value.

As long as you treat right-of-use assets as purchased equipment and treat the operating lease liability as a form of debt, your results will be theoretically consistent. Only operating profit requires an upward adjustment for implicit lease interest. Failing to adjust operating profit will undervalue equity, because implicit interest would be double-counted: once as part of lease expense and again as part of lease value when subtracting the value of leases from enterprise value to get to equity value.

Reorganizing the Financial Statements

To start the valuation of FlightCo, first reorganize the financial statements. Exhibit 22.3 presents the income statement, balance sheet, and statement of equity for FlightCo.

Using the information from FlightCo's financial statements, Exhibit 22.4 presents a calculation of NOPAT and its reconciliation to net income. The process starts by adding back the implicit interest embedded in the operating lease expense. To estimate implicit interest, multiply the prior year's operating lease liability by the interest rate used to value the operating lease. (If the company does not disclose the discount rate for operating leases in the notes, use the yield to maturity on AA-rated debt.) For FlightCo, embedded interest equals the operating lease liability of \$27.1 million multiplied by the interest rate of 5 percent. Estimate implicit interest using the operating lease liability and not the right-of-use asset.

EXHIBIT 22.3 **FlightCo: Financial Statements**

\$ million

	Year 0	Year 1	Year 2	Year 3
Income statement				
Revenue		75.0	75.0	75.0
Operating expenses		(40.0)	(40.0)	(40.0)
Operating lease expense ¹		(10.0)	(10.0)	(10.0)
Operating profit, unadjusted		25.0	25.0	25.0
Interest expense, debt ²		(0.4)	(0.3)	(0.3)
Earnings before taxes		24.6	24.7	24.7
Income taxes at 20%		(4.9)	(4.9)	(4.9)
Net income		19.7	19.7	19.8
Balance sheet				
Inventory	15.0	15.0	15.0	—
Right-of-use assets	27.1	18.5	9.4	—
Total assets	42.1	33.5	24.4	—
Operating leases	27.1	19.5	11.4	—
Debt	7.8	6.6	5.0	—
Equity	7.2	7.4	8.0	—
Liabilities and equity	42.1	33.5	24.4	—
Statement of equity				
Equity, start		7.2	7.4	8.0
Net income		19.7	19.7	19.8
Dividends and/or share repurchases		(19.5)	(19.1)	(27.8)
Equity, end		7.4	8.0	—

¹ Typically embedded in operating expenses, such as cost of sales.

² Interest equals 0.39, 0.33, and 0.25 in Year 1 through Year 3. As such, rounding errors affect earnings before taxes and net income.

EXHIBIT 22.4 **FlightCo: NOPAT and Reconciliation to Net Income**

\$ million

	Year 1	Year 2	Year 3
EBITA, ¹ unadjusted	25.0	25.0	25.0
Operating lease interest	1.4	1.0	0.6
EBITA, adjusted for lease interest	26.4	26.0	25.6
Operating taxes at 20%	(5.3)	(5.2)	(5.1)
NOPAT ²	21.1	20.8	20.5
Reconciliation to net income			
Net income	19.7	19.7	19.8
Interest expense, debt	0.4	0.3	0.3
Operating lease interest	1.4	1.0	0.6
Interest tax shield at 20%	(0.3)	(0.3)	(0.2)
NOPAT	21.1	20.8	20.5

¹ Earnings before interest, taxes, and amortization.² Net operating profit after taxes.

To calculate NOPAT, subtract operating taxes from adjusted operating profit. Operating taxes are estimated by multiplying operating profit by the operating tax rate. The resulting NOPAT for year 1 is \$21.1 million. The tax shield for embedded interest will be incorporated into the cost of capital.

We do not present a reorganized balance sheet for FlightCo, as the simplified balance sheet already matches invested capital. In general, include the right-of-use asset as part of invested capital and the operating lease liability as a source of financing.

Estimating Free Cash Flow

Once the financial statements are reorganized, estimate free cash flow. Exhibit 22.5 presents the free cash flow statement and its reconciliation to cash flow to investors for FlightCo. Free cash flow starts with NOPAT. Since FlightCo does not own property or equipment, there is no add-back for depreciation.⁴ From this value, subtract increases in working capital (inventory) and long-term assets (in this case, the right-of-use assets). Since both accounts are declining over time, they are positive numbers.

Note in Exhibit 22.5 how the summation of operating lease interest and the decrease in right-of-use assets equal the operating lease expense from Exhibit 22.3. Essentially, the valuation process eliminates the entire lease expense for existing assets from free cash flow. At the completion of the valuation, the lease of existing assets will not be valued as part of free cash flow, but rather as debt. Consistent with fundamental finance principles, this process separates investing flows from the manner in which they are financed.

⁴ To determine free cash flow, we incorporate the change in the right-of-use asset, which for FlightCo is decreasing over time. The positive value generated by the decline mimics the depreciation add-back. Essentially, the cash flows from capitalizing a lease are identical to those from purchasing an asset financed with debt.

EXHIBIT 22.5 **FlightCo: Free Cash Flow and Its Reconciliation**

\$ million

	Year 1	Year 2	Year 3
EBITA, ¹ unadjusted	25.0	25.0	25.0
→ Operating lease interest	1.4	1.0	0.6
EBITA, adjusted for lease interest	26.4	26.0	25.6
Operating taxes at 20%	(5.3)	(5.2)	(5.1)
NOPAT ²	21.1	20.8	20.5
Decrease (increase) in inventory	—	—	15.0
→ Decrease (increase) in right-of-use assets	8.6	9.0	9.4
Free cash flow	29.7	29.8	44.9
Interest tax shield at 20%	0.3	0.3	0.2
Cash flow available for investors	30.1	30.1	45.1
Reconciliation of free cash flow			
Interest, debt	0.4	0.3	0.3
→ Interest, operating leases	1.4	1.0	0.6
Decrease (increase) in debt	1.2	1.6	5.0
→ Decrease (increase) in operating leases	7.6	8.0	11.4
Flows to debt holders	10.6	10.9	17.3
Dividends	19.5	19.1	27.8
Cash flow to investors	30.1	30.1	45.1

¹ Earnings before interest, taxes, and amortization.² Net operating profit after taxes.

When reconciling cash flow to investors, treat embedded interest on operating leases and the change in the operating lease liability as a flow to debt holders. Again, note in Exhibit 22.5 how the summation of these two accounts matches the *cash-based* lease payment. Financing and its associated taxes should not be part of free cash flow.

Incorporating Operating Leases into Financial Projections

To forecast right-of-use assets, use the forecasting process introduced in Chapter 13. Link right-of-use assets to sales or a quantity-based measure, such as the number of units sold. In the airline industry, units are represented by number of available seat-miles. Make sure the mix of purchased and leased assets is consistent with the amount of capacity necessary to conduct operations.

Set the operating lease liability as a percentage of the right-of-use asset. While this estimation method is far from precise, flows to and from financing do not affect an enterprise-based valuation. Instead, financing affects valuation only through the target capital structure set in the weighted average cost of capital. If helpful, you can model the combination of operating leases and debt to the target capital structure, but it is not required.

Estimating the Cost of Capital

To discount free cash flow, use the weighted average cost of capital inclusive of the value of operating leases. Exhibit 22.6 presents the weighted average cost of capital for FlightCo.

We assume the company will maintain its current capital structure of 40 percent adjusted debt to value. Total debt equals the sum of the operating lease liability of \$27.1 million and traditional debt of \$7.8 million, divided by enterprise value, estimated at \$87.4 million. When estimating enterprise value, include operating leases as well. For FlightCo, the mix of operating leases and debt will change over time, but we set the combination to be stable at 40 percent of enterprise value. Since operating leases and interest expense are tax deductible, reduce the cost of capital for operating leases and debt by the company's marginal tax rate.

In the calculations for this example, the cost of equity is provided. In practice, the cost of equity must be estimated using beta. Following the principles of Chapter 15, start by estimating an industry unlevered beta. Unlever each company's beta using a debt-to-equity ratio adjusted for operating leases. Next, to determine the target company's beta, relever the unlevered beta to the target company's capital structure, again inclusive of leases.

Moving from Enterprise Value to Equity Value

To calculate enterprise value, discount free cash flow at the weighted average cost of capital, both inclusive of leases. Exhibit 22.7 shows the enterprise DCF valuation for FlightCo.

Since free cash flow excludes future payments related to existing operating leases, the value of operating leases must be deducted from enterprise value to determine intrinsic equity value. For FlightCo, enterprise value is estimated at \$87.4 million. Deducting the present value of the operating lease liability (\$27.1 million) and debt of \$7.8 million leads to an equity value of \$52.4 million.

EXHIBIT 22.6 **FlightCo: Weighted Average Cost of Capital (WACC)**

%						
Source of capital	Value, \$ million	Proportion of total capital	Cost of capital	Marginal tax rate	After-tax cost of capital	Contribution to weighted average
Operating leases ¹	27.1	31.0	5.0	20.0	4.0	1.2
Debt	7.8	9.0	5.0	20.0	4.0	0.4
Equity	52.4	60.0	12.0		12.0	7.2
WACC	87.4	100.0				8.8

¹ The present value of operating leases, found in the liabilities section of the balance sheet.

EXHIBIT 22.7 **FlightCo: Enterprise DCF Valuation**

\$ million, except where noted

Forecast year	Free cash flow (FCF)	Discount factor at 8.8%	Present value of FCF
Year 1	29.7	0.919	27.3
Year 2	29.8	0.845	25.2
Year 3	44.9	0.776	34.9
Value of operations			87.4
Less: Operating leases ¹			(27.1)
Less: Debt			(7.8)
Equity value			52.4

¹ The present value of operating leases, found in the liabilities section of the balance sheet.**Valuation Using Cash Flow to Equity**

In general, we do not recommend a valuation model based on cash flow to equity, because it mixes assets of different risks and commingles operating performance with the capital structure. If implemented properly, however, a cash-flow-to-equity valuation can confirm the accuracy of the enterprise DCF process described in this chapter. It can also provide insight into choices made during the capitalization process.

Exhibit 22.8 presents cash flow to equity for FlightCo. In this exhibit, each line item represents actual cash flowing into or out of the company, from the equity holder's perspective. In the equity model, do not capitalize lease expense. Instead, deduct the cash payments paid to the lessor when they occur. Since leases are expensed and not capitalized, do not include either the change in the right-of-use asset or the change in the operating lease liability. This stands in contrast to debt flows, where both interest expense and payoff of debt are included in the calculation, since they represent actual cash flows.

EXHIBIT 22.8 **FlightCo: Cash Flow to Equity Holders**

\$ million

	Year 1	Year 2	Year 3
Revenue	75.0	75.0	75.0
Operating costs	(40.0)	(40.0)	(40.0)
Lease payments ¹	(9.0)	(9.0)	(12.0)
Interest expense, debt	(0.4)	(0.3)	(0.3)
Earnings before taxes	25.6	25.7	22.7
Income taxes	(4.9)	(4.9)	(4.9)
Earnings after taxes	20.7	20.7	17.8
Change in inventory	0.0	0.0	15.0
Increase (decrease) in debt	(1.2)	(1.6)	(5.0)
Cash flow to equity	19.5	19.1	27.8

¹ Cash-based lease payments.

EXHIBIT 22.9 **FlightCo: Valuation Using Cash Flow to Equity**

\$ million, except where noted

Forecast year	Cash flow to equity (CFE)	Discount factor, at 12%	Present value of CFE
Year 1	19.5	0.893	17.4
Year 2	19.1	0.797	15.3
Year 3	27.8	0.712	19.8
Equity value			52.4

Exhibit 22.9 values cash flow to equity at the cost of equity. The cost of equity used to discount equity cash flows equals the cost of equity used to determine the weighted average cost of capital. One may think the cost of equity should fall, since the leverage associated with operating leases is being ignored. This is not the case, however. The underlying risk of equity has not changed when switching models, so the cost of equity should not change either.

Discounting cash flow to equity at a 12 percent cost of equity leads to an equity valuation of \$52.4 million. This is the same valuation as we calculated by using the enterprise DCF model.

ADJUSTING HISTORICAL FINANCIAL STATEMENTS FOR OPERATING LEASES

As time progresses, distortions caused by operating leases will be forgotten in the same way most investors have forgotten the adjustments required for the long-defunct pooling of interests prior to 2000. Until then, it is important to recognize that historical financial statements will remain unadjusted. To assure consistency in historical analysis prior to 2019, adjust historical statements to match current accounting policy.

In the analysis of Costco presented in Chapter 11 and Appendix H, we used information from their annual report to value operating leases and adjust the historical financial statements accordingly. Exhibit 22.10 presents the valuation of operating leases for Costco in 2019, the year before Costco adopted the new standard.⁵ To value operating leases, we discount future lease payments at the cost of AA-rated debt. Lease commitments are reported in note 5 of Costco's 2019 annual report. The company reports only the first five years of lease payments year by year. Lease payments beyond 2024 are lumped into a single undiscounted number. At the bottom of Exhibit 22.10, we value the lump sum using an annuity formula. In the formula, set the cash

⁵ Companies whose fiscal year ends after December 15 had to implement the new leasing standard in 2019. Since Costco's fiscal year ended on September 1, 2019, it chose to adopt the new standard in 2020.

EXHIBIT 22.10 **Costco: Operating Lease Valuation, 2019**

\$ million

Forecast year	Rental commitments	Discount factor at 3.6% ¹	Present value of payments
2020	239.0	0.965	230.6
2021	229.0	0.931	213.2
2022	202.0	0.898	181.5
2023	193.0	0.867	167.3
2024	181.0	0.837	151.4
Payments beyond 2024	1,757.1	0.837	1,470.0
Value of operating leases			<u>2,414.0</u>
Value beyond 2024			
Rental commitments beyond 2024	2,206.0		
/ Final year rental payment	181.0		
= Number of years	12.19		

Annuity value of \$181.0 per year for 12.19 years = \$1,757.1

¹ Yield-to-maturity on 10-year AA-rated debt.

Source: Costco 2019 annual report, note 5.

flow equal to the 2024 lease payment. To estimate the number of years, divide the undiscounted lump sum by the 2024 rental payment. For Costco, the annuity value equals almost \$1.8 billion. Since the annuity values the lump-sum payments beyond 2024 as of 2024, make sure to discount the result back to 2019, as you would any other cash flow.

In general, when reorganizing the balance sheet, include the value of operating leases as part of invested capital. Incorporate the corresponding liability as a debt equivalent. For past statements, the asset will equal the liability.

In Chapter 11, we adjust Costco's reorganized financial statements for operating leases. Exhibit 11.5 presents invested capital inclusive of operating leases, and Exhibit 11.9 adjusts EBITA and NOPAT for implicit interest. To calculate implicit interest for 2019, multiply the cost of debt of 3.63 percent by the 2018 capitalized operating lease of \$2.5 billion. The resulting adjustment to EBITA equals \$91 million.

AN ALTERNATIVE METHOD FOR VALUING OPERATING LEASES

To capitalize operating leases on the balance sheet, the company discounts future lease commitments at the company's borrowing rate. For short-term leases, this methodology will understate the actual value of the asset, since it ignores the residual value of the asset being returned to the lessor. Consider FlightCo, which rented an aircraft for three years of the plane's 40-year life. A new aircraft may cost \$125 million, but three years of rental expense will be far lower.

While using the present value of lease payments in place of the true asset value will not bias the valuation, it will understate the value of the assets being deployed to run operations. (The error will be largest for short-term leases on long-term assets. In the case of finance leases, the error will be small, since the lease life more closely matches the asset life.) When benchmarking two companies, one that purchases assets and one that rents them, the comparison will not be like-for-like, even under new accounting standards.⁶ Distortions to ROIC and capital turnover will be largest when leased assets are a significant proportion of invested capital.

One way to create a like-for-like comparison for companies with different leasing policies is to estimate each company's asset value by using a perpetuity. To see how, let's examine the determinants of rental expense. To compensate the lessor properly, the rental expense includes compensation for the cost of financing the asset (at the cost of secured debt, denoted by k_d in the following equations) and the periodic depreciation of the asset (for which we assume straight-line depreciation). The following equation solves for periodic rental expense:

$$\text{Lease Expense}_t = \text{Asset Value}_{t-1} \left(k_d + \frac{1}{\text{Asset Life}} \right) \quad (22.1)$$

To estimate the asset's value, rearrange equation 22.1 as follows:

$$\text{Asset Value}_{t-1} = \frac{\text{Lease Expense}_t}{\left(k_d + \frac{1}{\text{Asset Life}} \right)} \quad (22.1)$$

Lease expense is disclosed in the notes, and the cost of debt can be estimated using AA-rated yields. This leaves only the asset life, which is often unreported. If this is the case, search the notes for the type of asset being leased, and estimate an asset life appropriate to the asset type. As an alternative, Lim, Mann, and Mihov propose using property, plant, and equipment (PP&E) divided by annual depreciation.⁷ In their research, they examined 7,000 firms over 20 years and computed the median asset life at 10.9 years.

⁶In this section, we focus on the distortions to benchmarking caused by different leasing policies. Companies choose different leasing policies for many reasons, including flexibility and taxes.

⁷S. C. Lim, S. C. Mann, and V. T. Mihov, "Market Evaluation of Off-Balance Sheet Financing: You Can Run but You Can't Hide" (EFMA 2004 Basel Meetings paper, European Financial Management Association, December 1, 2003).

CLOSING THOUGHTS

Recent changes in lease accounting have brought financial reporting very close to the core principles of this book. Still, a proper valuation requires special care regarding operating leases. To inform better forecasts, adjust statements created prior to the accounting rules changes to incorporate operating leases. If you do not, apples-to-oranges comparisons may obscure crucial trends. For statements reported after the changes, remember to eliminate any interest embedded in operating expenses. Otherwise, you risk double-counting embedded interest, biasing your valuation downward. The task requires discipline and attention to detail, but with practice, the adjustments should become routine.

REVIEW QUESTIONS

1. What do the 2019 U.S. GAAP rules say about how to treat operating and finance leases? How do these rules differ from IFRS rules?
2. PharmaCo, a chain of pharmacies, leases a significant portion of its assets. Their reported financial statements are shown in Exhibit 22.11. PharmaCo has lease payments over the next three years of \$7.0, \$8.0, and \$10.0 million. Using Exhibits 22.4 and 22.5 as a guide, estimate PharmaCo's NOPAT and FCF, assuming a cost of debt of 5 percent.

EXHIBIT 22.11 **PharmaCo Financial Statements**

\$ millions

	Year 0	Year 1	Year 2	Year 3
Revenue		100.0	100.0	100.0
Operating expenses		(60.0)	(60.0)	(60.0)
Operating lease expense ¹		(8.3)	(8.3)	(8.3)
Operating profit, unadjusted		31.7	31.7	31.7
Interest expense, debt		(0.3)	(0.2)	(0.1)
Earnings before taxes		31.4	31.5	31.5
Income taxes at 20%		(6.3)	(6.3)	(6.3)
Net income		25.1	25.2	25.2
Balance Sheet:				
Inventory	12.0	12.0	12.0	-
Right-of-use assets	22.6	15.4	7.9	-
Total assets	34.6	27.4	19.9	-
Operating leases	22.6	16.7	9.5	-
Debt	5.3	3.8	2.9	-
Equity	6.7	6.8	7.4	-
Liabilities and equity	34.6	27.4	19.9	-
Statement of Equity:				
Equity, start		6.7	6.8	7.4
Net income		25.1	25.2	25.2
Dividends and/or share repurchases		(25.0)	(24.6)	(32.6)
Equity, end		6.8	7.4	-

¹ Typically embedded in operating expenses.

3. Using Exhibits 22.6 and 22.7 as a guide, estimate PharmaCo's WACC and enterprise DCF valuation, assuming a cost of equity of 12 percent.
4. Using Exhibits 22.8 and 22.9 as a guide, estimate PharmaCo's cash flow to equity holders and valuation using cash flow to equity.
5. CashCo Industries leases a significant portion of its assets, expecting \$25 million in rental expense next year. CashCo Industries can borrow at 6 percent, and the average life of leased assets is seven years. CashCo's largest competitor, ChargeCo, owns all of its assets. For a more direct comparison of ROIC and capital turnover figures for CashCo and ChargeCo, estimate the value of leased assets outlined in Equation (22.2). If you misestimate the average life to be ten years, how large will the valuation error be?
6. Many financial analysts estimate the value of operating leases by discounting rental payments provided in the annual report at the cost of debt. Is this method likely to overestimate or underestimate the value of leased assets? Why?

